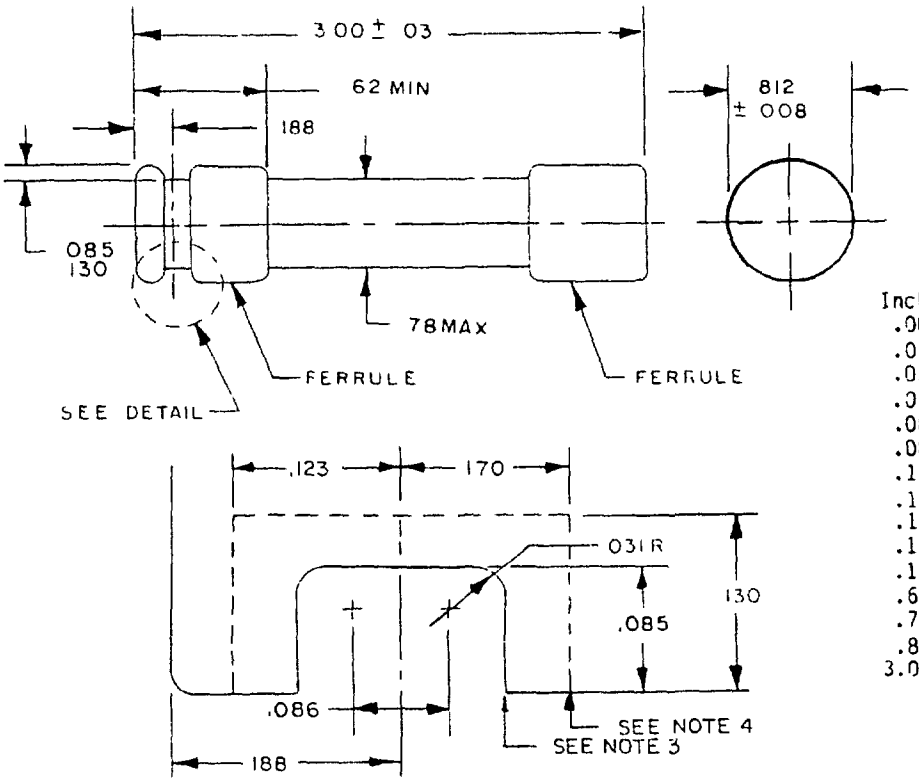
FEDERAL SPECIFICATION SHEET

FUSES, CARTRIDGE, HIGH-INTERRUPTING CAPACITY, CLASSES RK1 AND RK5 (CURRENT-LIMITING) 250 VOLTS, 31-60 AMPERES

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

The complete requirements for procuring the fuses described herein shall consist of this document and the latest issue in effect of specification W-F-1814/GEN.



REJECTION GROOVE DETAIL

Inches	mm
.008	0.20
.03	0.76
.031	0.79
.070	1.78
.085	2.15
.086	2.18
.123	3.12
.130	3.30
.150	3.81
.170	4 32
.188	4.78
.62	15 8
.78	19.8
.812	20 62
3.00	76 2

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only
3. Solid line indicates location, shape, and dimensions for minimum rejection groove
4. Dashed line indicates location, shape, and dimensions for maximum rejection groove

FIGURE 1 Classes RK1 and RK5 fuses, 250 volts, 31-60 amperes

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited

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REQUIREMENTS

Design and construction See figure 1.

Physical. Nonrenewable.

Terminals Ferrule type. The ferrules shall be approximately circular in cross-section

Material Brass, copper, or copper alloy.

Strength. 5 inch-pound torque between body and ferrules.

Alignment The inside diameter of tubular gauge shall be not more than 0.823 inch.

Body

Insulating material. Fiber, ceramic, melamine-impregnated glass fiber, or other suitable material.

Electrical

Voltage 250 volts or less.

Frequency 48 to 60 hertz.

Current See table I.

Current carrying capacity 110 percent indefinitely with temperature rise not to exceed 50°C (90°F) above ambient on the body and the ferrules.

Overload interrupt

Instantaneous Shall interrupt within 1 hour at 135 percent of rated current and within 4 minutes at 200 percent of rated current.

Time delay In addition to the above requirement, the fuse shall not interrupt 500 percent of rated current within 10 seconds

Interrupting capacity rating 200,000 amperes rms symmetrical at 250 volts, 48 to 60 hertz and a power factor of 20 percent or less. Closing angle shall be essentially at the zero of the voltage wave (maximum offset) or later, to produce start of arcing within 30 electrical degrees prior to system peak voltage

Threshold ratio 30 maximum for RK1; 65 maximum for RK5.

Peak let-thru current See table II.

Maximum clearing I^2T See table II.

Applicable fuseholder Part number W-F-870/7-02 of W-F-870/7

Government part number The government part number shall consist of the prefix "WF1814/57-" followed by the part number designation shown in table I.

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TABLE I. Current rating and part number designation

Current in amperes	Part number designation			
	RK5		RK1	
	Instantaneous	Time delay	Instanta-neous	Time delay
35	F535	D535	F135	D135
40	F540	D540	F140	D140
45	F545	D545	F145	D145
50	F550	D550	F150	D150
55	F555	D555	F155	D155
60	F560	D560	F160	D160

TABLE II Peak let-thru and I^2T current.

	Between threshold and 50 kA		100 kA		200 kA	
	RK1	RK5	RK1	RK5	RK1	RK5
Maximum acceptable peak let-thru current ($I_p \times 10^3$)	10	20	12	21	15	25
Maximum clearing I^2T (amperes squared seconds) ($I^2T \times 10^3$)	40	200	40	200	50	200

MILITARY INTERESTS

Custodians

Army - ER
Navy - YD
Air Force - 85

Review Activities

Army - AL, AT, ME
Air Force - 99
DLA - ES
NS

User Activities

Army - AR
Navy - OS

CIVIL AGENCY COORDINATING ACTIVITIES

DOE - DOE
COMMERCE - NBS
DOT - ACO, APM
GPO - GPO
GSA - FSS
JUSTICE - FPI
NASA - JFK, LRC
USPS - POS
VA - OSS

PREPARING ACTIVITY

DLA - ES

(Project 5920-0434-02)

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